



Cold Spring Harbor Laboratory
P.O. Box 100, 1 Bungtown Road
Cold Spring Harbor, New York 11724

Justin B. Kinney, Ph.D.
Professor and Chair
Simons Center for Quantitative Biology
Cold Spring Harbor Laboratory
(631) 897-5731
jkinney@cshl.edu

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Subject: Manuscript submission to GENETICS

Dear Editor,

I kindly ask that you consider my manuscript, “*Mount Fuji’s stubby peak: the genotypic density of additive landscapes near maximal fitness*,” for publication in the Special Issue on Fitness Landscapes.

In this work I address a fundamental question about fitness landscapes: how many high-fitness genotypes are there at a given fitness value? Even for the simplest and most widely used class of fitness landscapes—additive landscapes (a.k.a. Mount Fuji landscapes)—the density of genotypes near maximal fitness remains unknown. Here I answer this question for both additive and globally epistatic fitness landscapes.

I begin by deriving a general mathematical expression that describes how the genotypic density of additive landscapes behaves in the near-peak regime. This expression takes the form of an exponentiated power law, where the specific value of the exponent depends on the quantitative details of the landscape. Among other things, this expression reveals that Mount Fuji landscapes have tops that are more akin to gently rolling hills than to sharply peaked summits.

I show that this mathematical expression accurately describes the densities of simulated and empirical fitness landscapes. I then extend these findings to a large class of globally epistatic landscapes, i.e., landscapes defined by nonlinear functions of one or more additive traits.

To my knowledge, these results provide the first quantitative description of near-peak genotype abundance for any class of realistic fitness landscape. In addition to the calculations of genotypic density that this work enables, the mathematical methods I use can likely be extended to related problems, such as how the distribution of fitness effects (DFE) behaves near fitness peaks, or how Fisher’s geometric model is affected by genotypic density. I therefore believe this work fits squarely within the scope of this Special Issue as a new theoretical contribution to the study of fitness landscapes.

I suggest the following experts as potential reviewers:

- Richard Neher, University of Basel, richard.neher@unibas.ch
- Armita Nourmohammad, Yale University, armita.nourmohammad@yale.edu
- Ville Mustonen, University of Helsinki, ville.mustonen@helsinki.fi
- Gašper Tkačik, ISTA, gtkacik@ist.ac.at
- Joachim Krug, University of Cologne, jkrug@uni-koeln.de

Sincerely,

Justin B. Kinney, PhD